Bitcoin's volatility and trading intensity in 10 figures

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General context

Bitcoin, introduced in 2009 by the pseudonymous Satoshi Nakamoto, is often interpreted as a reaction to the 2008 financial crisis and the resulting bank bailouts. It proposes a decentralized monetary system where no central authority controls issuance or transactions, relying instead on a distributed network.

Over time, Bitcoin's price and ecosystem have exhibited extreme volatility, with repeated cycles of rapid growth and sharp declines. These boom-bust phases have also been mirrored in the development and collapse of infrastructure such as exchanges, mining operations, and financial services around Bitcoin.

Executive summary

Bitcoin exhibited extreme price growth, far outperforming traditional assets in nominal and real terms, with inflation having only a limited relative impact. At the same time, it showed on average substantially higher volatility than equities, gold, and oil.

Despite weak to moderate correlations with other assets, these are insufficient to offset its dominant volatility contribution. Thus, adding Bitcoin to a portfolio is unlikely to reduce the overall volatility.

Historical price development

Figure 1 shows the price development of bitcoin over the years. The years 2009-2012 are missing due to data availability. The goal of this figure is to visualize the many bullish and bearish phases of the market.

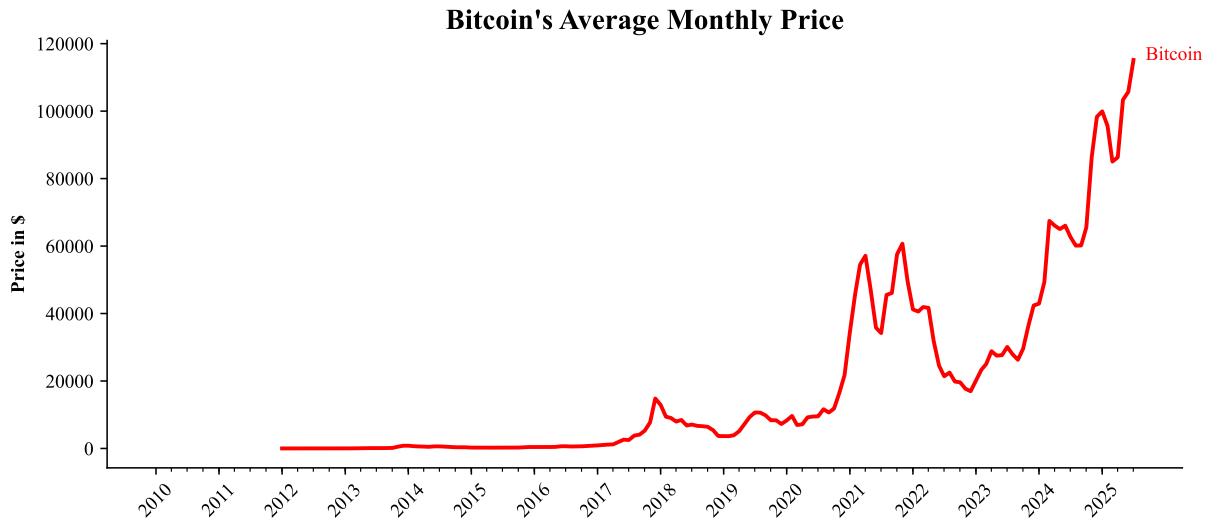

Figure 1

Most important developments:

- The first popularity wave in 2017
- Bulling market and update of the all-time-high at the beginning of 2021
- 67% loss between the all-time-high and beginning of 2023
- The strongest bull market yet, with Bitcoin crossing $100.000 a coin at the beginning of 2025

Figure 2 shows the comparison of Bitcoin's price development compared to other assets. The charts how log price index of the assets have developed (normalized to 1 on 1st of January 2012). This was done for 2 reasons:

1. Normalize data to visualize what return an investor would have achieved, had they invested in each of the assets at the beginning of 2012
2. Due to explosive growth of Bitcoin, the log index was used to keep the chart readable.

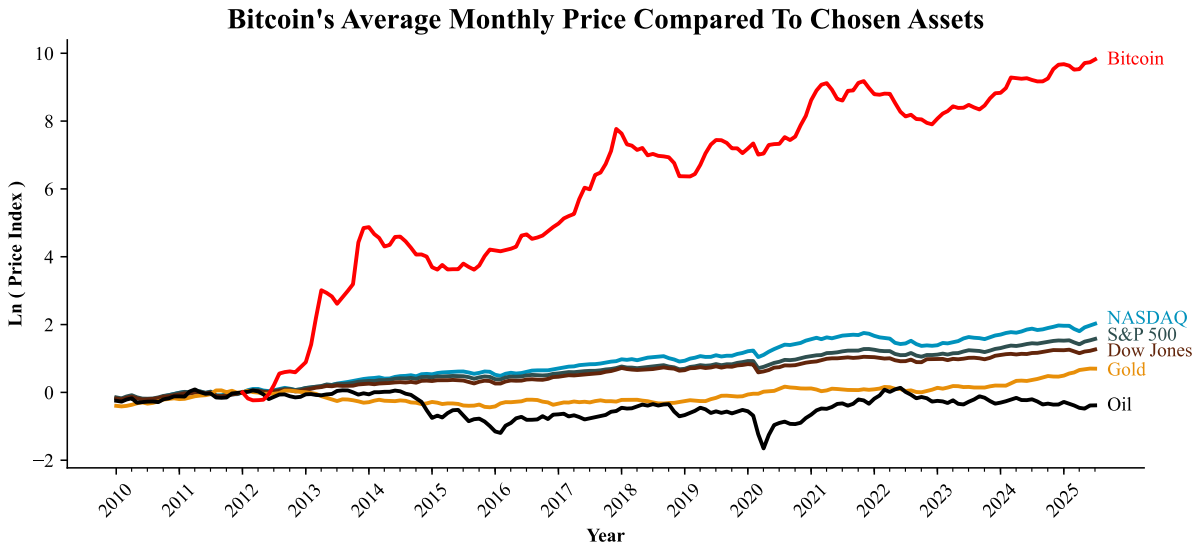

Figure 2

From the chart we can derive 2 things:

- Bitcoin's volatility as can be seen in the *Figure 1*
- Near constant log returns for other assets excluding oil

The interpretation of the Y-axis is relatively complicated and is not the focus of the figure. You can still find it in the notebook under `phase1/analysis.ipynb` in the `Plot data / Summary` section.

Separately from the visualization, we note that Bitcoin's returns shadow any asset in a similar timeframe with whopping 1,702,630.84% growth over the last ~13 years. Such comparisons are however very naive since Bitcoin was founded in 2009, which makes it, in the grand scheme of things, a brand-new investment tool. Remarkably, Bitcoin's growth over the years averaged arithmetically at about ~5,38% a month. While other assets have experiences more stable arithmetic average growth rates of about ~0.8% a month, they once again did not come close to Bitcoin's return. Past returns however do not guarantee future growth especially amidst increased global uncertainty and regulation.

Inflation adjusted returns

The 15 years between 2010 and 2025 have also experiences a certain level of inflation albeit a historically very low one.

The *Figure 3* shows Bitcoin's price development in nominal and inflation adjusted terms. The CPI is normalized to 1 in January 2010. There we can see that the inflation had only a minor effect on Bitcoin's real returns before 2020. After 2020 it has become a sizable factor in the real return of the cryptocurrency. The latter has however outpaced inflation and gave its investors positive returns in real terms.

Figure 4 is used for a reference to show that the real return for more traditional assets was reduced more visibly by inflation compared to Bitcoin with the reason being the staggering growth of the cryptocurrency.

Nominal vs real price developments

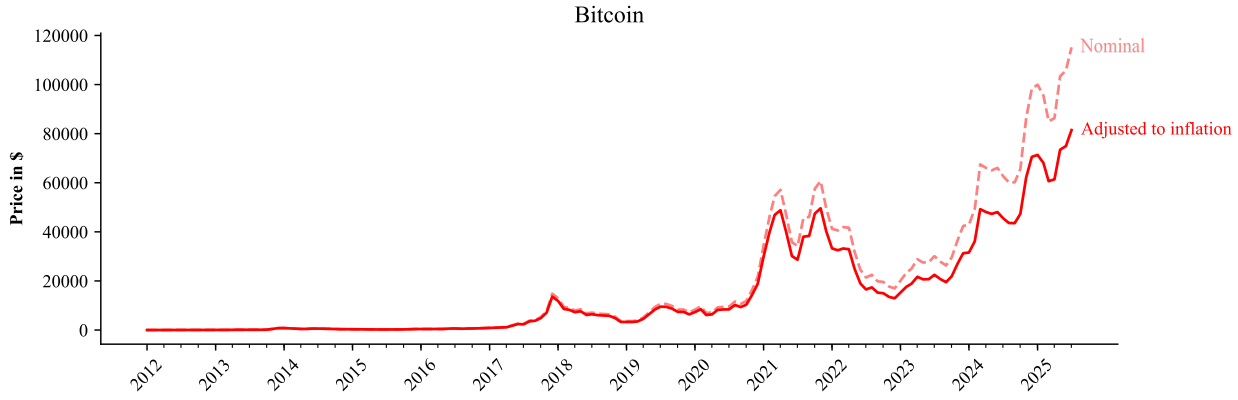

Figure 3

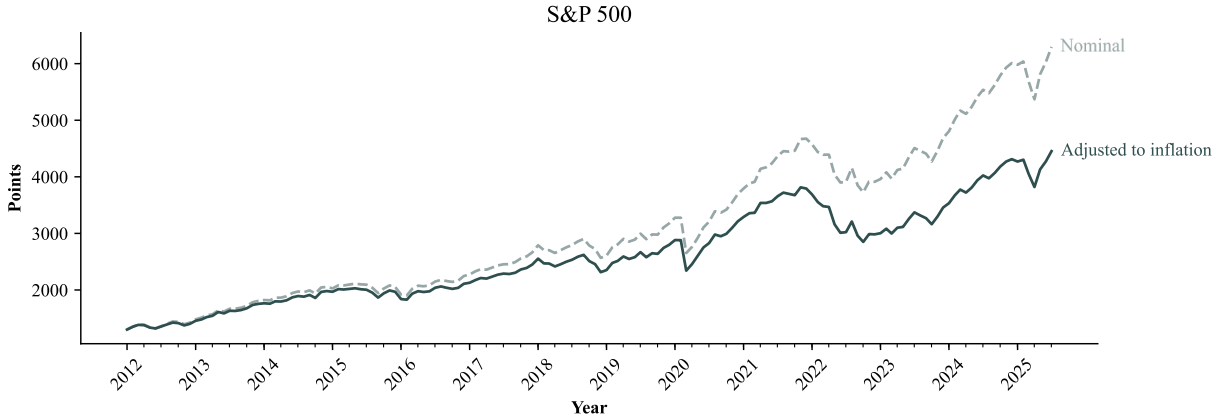

Figure 4

Volatility

The incredible returns however come with drawbacks, one of which is volatility. *Figure 5* depicts standard deviation (volatility) of assets' daily log return (based on average daily close) aggregated for each year.

Bitcoin's volatility is visually much more comparable to oil than to any other asset, with most assets' daily volatility visually lying at around 1 pp. Interestingly, as the time went on and Bitcoin's popularity grew it seems to correlate with sinking volatility. One of possible reasons why this has happened may be the fact that market became more heterogenous.

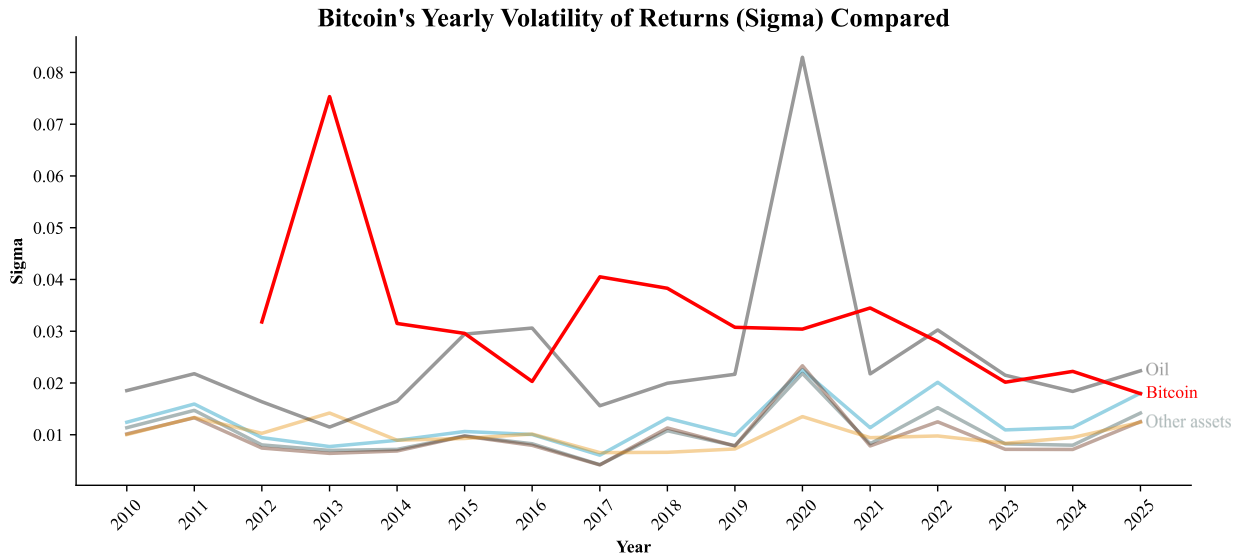

Figure 5

Having visualize the daily volatility, we will now formally test monthly price volatility. The reason for this frequency reduction is explained further down in the analysis. Hence, *Figure 5* and the test capture different types of volatility, which are not directly comparable. The reason for this is the fact that the analyses measure different types of volatility (daily vs. monthly).

To formally test Bitcoin's volatility against other assets the following test setup has been chosen:

- Paired bootstrap variance ratio

$$R_i = \frac{\sigma_{BTC}^2}{\sigma_i^2} \; with \; i \; \in \{NASDAQ, \quad S\&P\,500, \quad Dow\,Jones, \quad Gold, \quad Oil\}$$

- Is tested at $\alpha = 0.01$ using the method of confidence intervals
- Where σ_{it} is the standard deviation of log returns based on monthly average close of each asset.
- With

$$H_o: R_i \leq 1$$
$$H_1: R_i > 1$$

- Pooled over all known Bitcoin market regimes. Thus, the variance ratio mixes all market structures. Hence, this likely introduces upward bias of the CI compared to a hypothetical stationary single-regime parameter.
- 200.000 resamples with BCa as method to find quantiles

Results:

	Lower CI	Upper CI	SE
S&P 500	16.31	Inf.	15.7
NASDAQ	14.84	Inf.	8.08
Dow Jones	14.02	Inf.	18.02
Gold	28.2	Inf.	10.52
Oil	2.16	Inf.	1.5

- i.i.d assumption required for validity of the bootstrap test is likely violated, due to volatility clustering and structural changed in the financial markets. However, by aggregating data to monthly basis, we partially offset the short-term dependencies.
- Therefore, the confidence intervals should be interpreted as approximate and descriptive, with understanding that the remaining dependence will bias the confidence intervals and pooling over all regimes makes the estimate unsuitable for further inference.

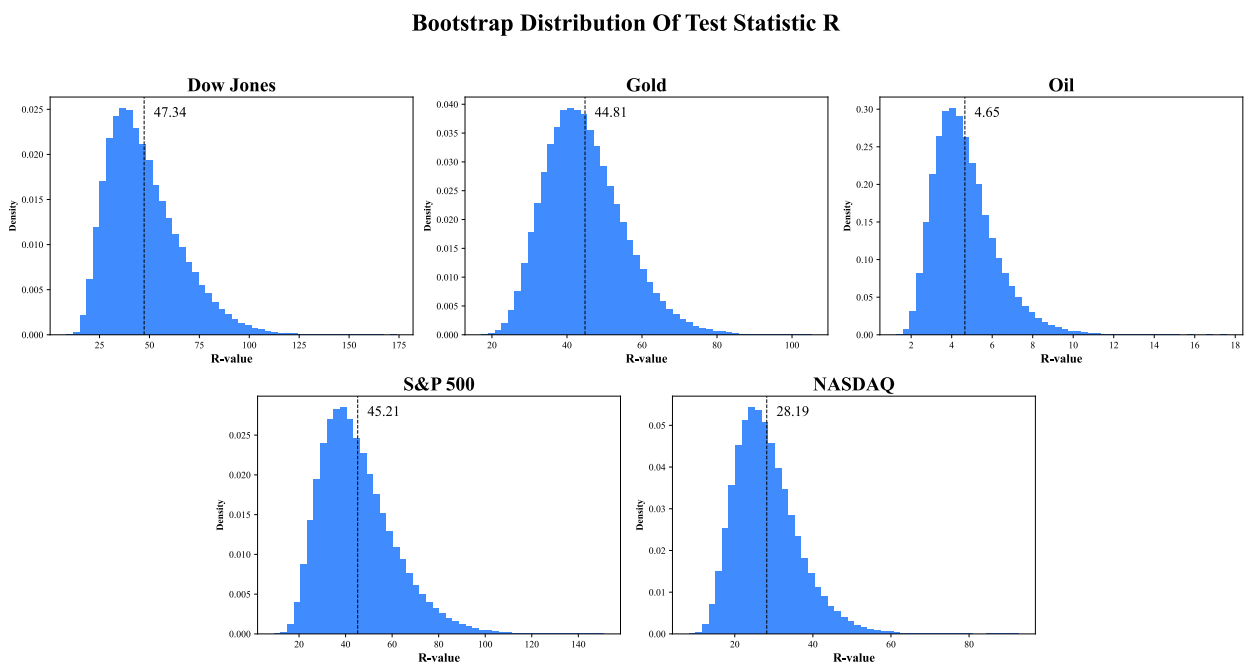

Figure 6

Figure 6 is required for visual confirmation of the validity of the bootstrap test. The main things to look out for is well behaved distribution with one peak. The grey line shows mean of the resampled R-statistic.

Interpretation:

- For each asset H_o is rejected at $\alpha = 0.01$
- Thus R-statistics is significantly bigger than 1

We take a square root of the R-Statistics' confidence interval to get volatility ratios. Hence, roughly speaking, Bitcoin is at least:

- ~4 times more volatile as NASDAQ
- ~1.4 times more volatile than oil

- ~5 times more volatile than gold
- ~4 times more volatile than S&P 500
- ~3.5 times more volatile than Dow Jones

in log returns based on 99% confidence interval. This is not necessarily the textbook interpretation but is very helpful to understand the magnitude of Bitcoin's volatility.

Correlation

Figure 7 shows Pearson's linear correlation coefficients, where 1 indicates perfect positive correlation, -1 perfect negative correlation, and 0 no linear relationship.

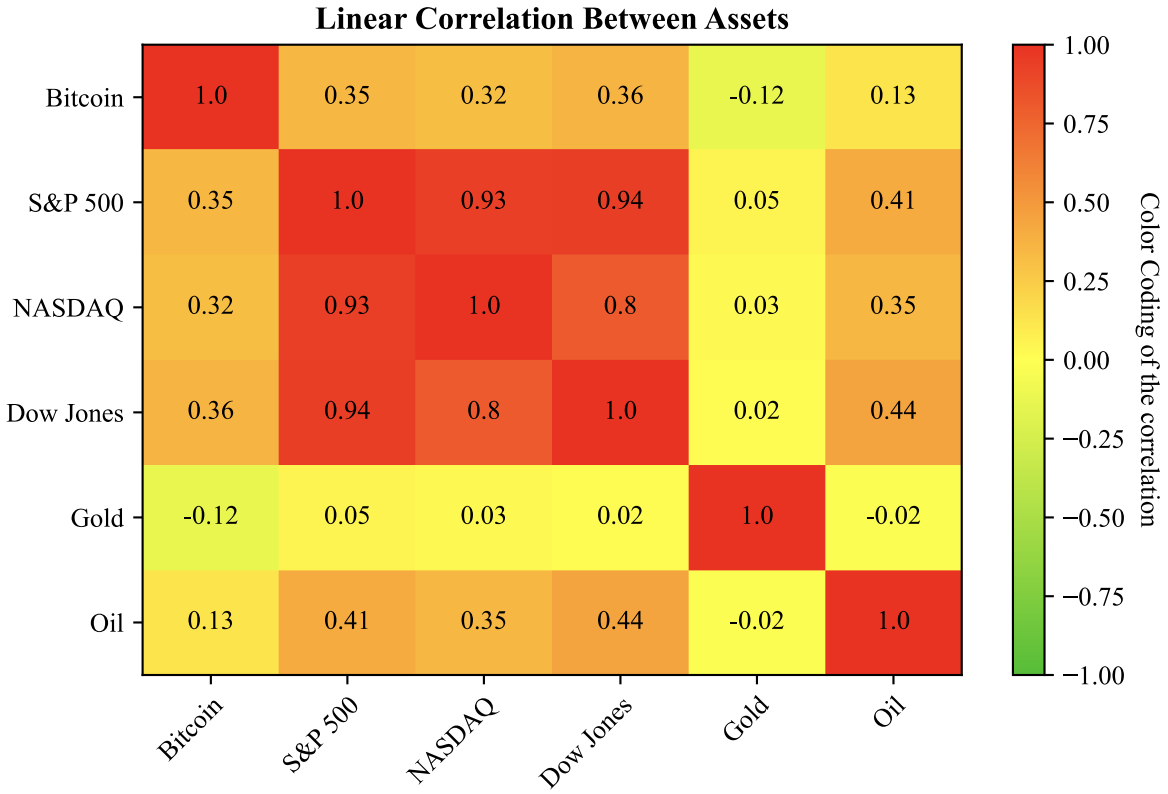

Figure 7

While Bitcoin has exhibited weak to slightly negative correlation with some traditional assets on average[1], this property alone is probably not sufficient to offset its substantially higher volatility. In principle, low or negative correlation can provide diversification benefits; however, given Bitcoin's magnitude of variance relative to other assets, it is unlikely that its inclusion would reduce overall portfolio volatility under typical conditions. Thus, correlation may appear as a redeeming quality, but in this case, it is dominated by Bitcoin's extreme volatility.

[1] Over all market regimes

Trading intensity

Results

Lastly, we will compare Bitcoin's $Trading\ intensity_i = \frac{Trading\ volume_i}{Outstanding\ supply_i}$. This can be roughly interpreted as a turnover rate. This metric is however explicitly not defined as turnover rate due to underlying data.

Figure 8 shows us raw trading intensities. As you may notice oil overshadows all the assets by a long shot. The reduced volatility before 2013 comes from missing data from NYMEX. The missing data was extrapolated based on the mean volume from NYMEX + valid Shanghai and TOCOM data. The same goes for gold before 2013.

Figure 9 captures Bitcoin's trading intensity without oil. Here we see the true magnitude of Bitcoin's trading intensity.

Figure 10 uses log scale to put all the assets into a comparable context. The goal of this figure is to show the relatively constant levels of trading intensity with oil leading the pack and ETFs being the least trade intensive ones. This figure is not meant to be comparable to other ones and is used only for level comparisons.

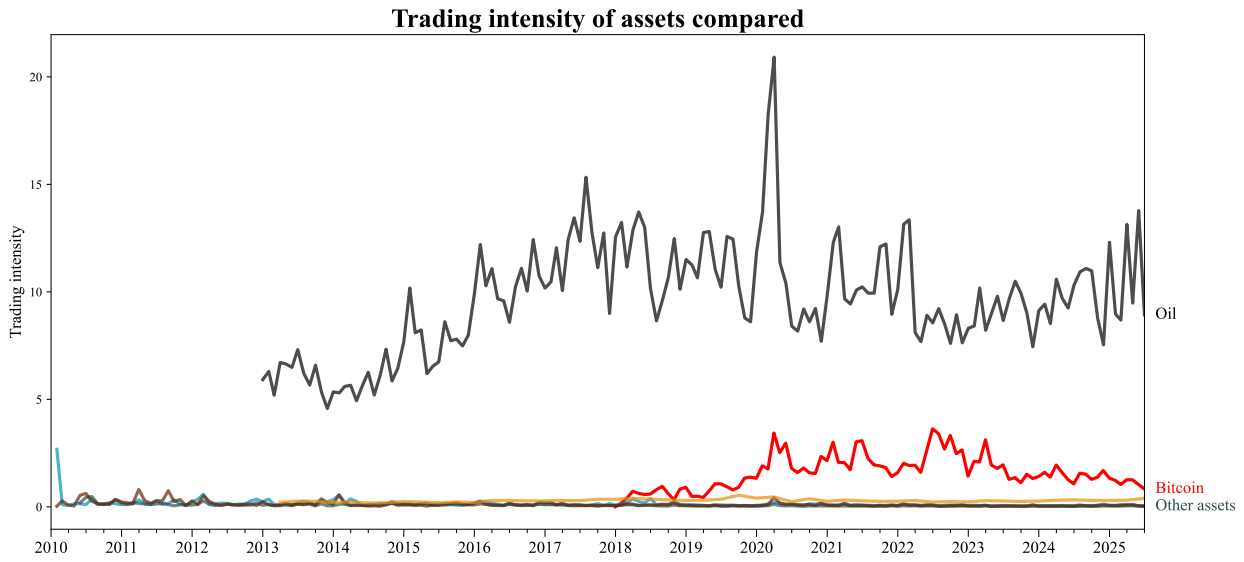

Figure 8

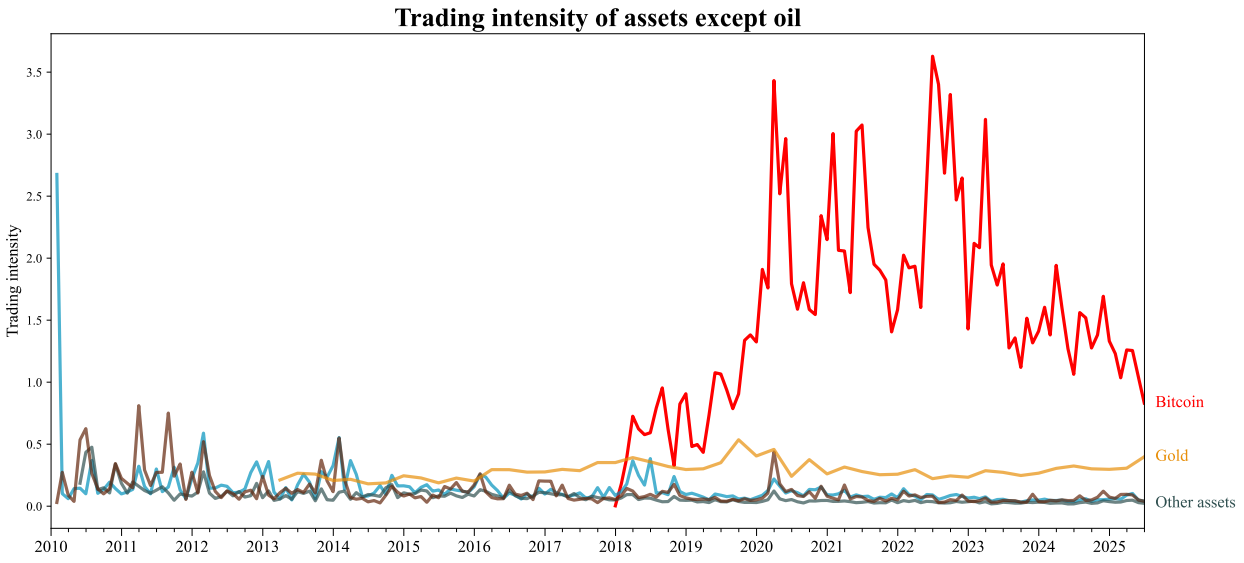

Figure 9

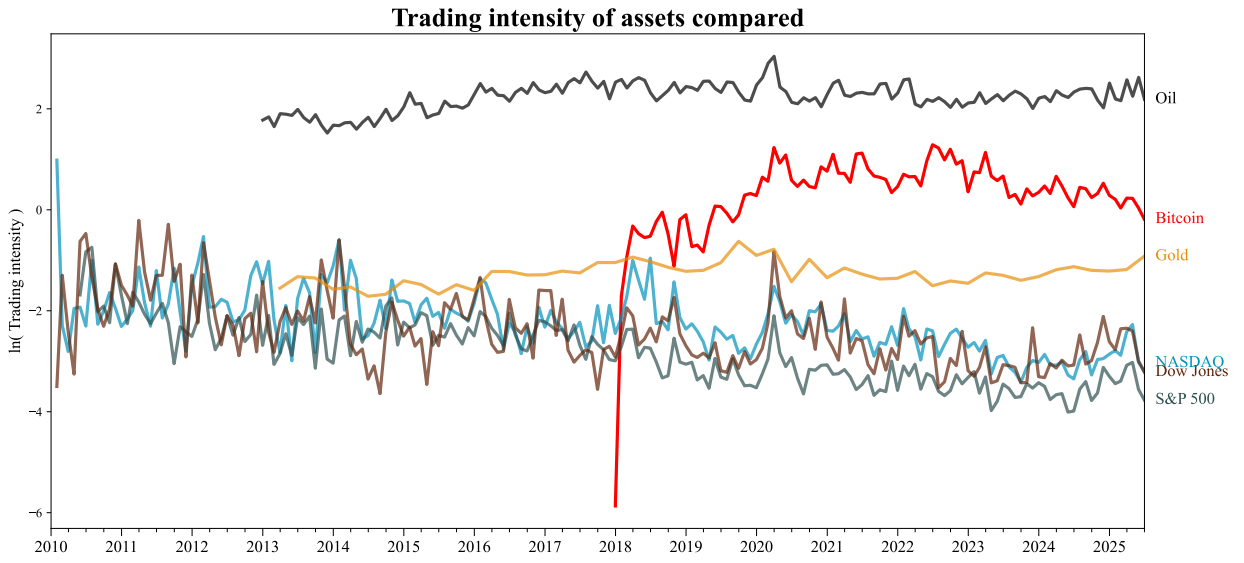

Figure 10

Methodology

Proxys for supply:

Name	**Proxy / proxies**
Indices	ETFs in circulation
Gold	Total gold above ground
Oil	Total production in a period
Bitcoin	Coins in circulation

Why: Since oil is a one-time consumption item, we assume its total supply to be 0 after a period.

Proxys for trading volume:

Name	**Proxy / proxies**
Indices	Futures
Commodities	Physical delivery futures
Bitcoin	Fiat settled futures

Why:

a) Why Bitcoin futures exactly, why not spot – to incorporate short cells
b) It's easier to capture direct futures volume, no need to search for each derivative of an asset and then build a separate pipeline to fetch it
c) Investors speculate of "consequential" futures (with a delivery) less. Thus, giving a speculation picture, which excludes small day-traders who bump up volume and may not be present in other asset classes

Target granularity:

<table>
<tr><th>Metric
Asset</th><th>Supply</th><th>Trading volume</th></tr>
<tr><td>S&P 500</td><td colspan="2" rowspan="3">Monthly</td></tr>
<tr><td>NASDAQ</td></tr>
<tr><td>Dow Jones</td></tr>
<tr><td>Gold</td><td rowspan="2">Monthly (fallback quarterly)</td><td rowspan="2">Monthly (fallback quarterly)</td></tr>
<tr><td>Oil</td></tr>
<tr><td>Bitcoin</td><td colspan="2">Monthly</td></tr>
</table>

Assumptions and important notes

- Bitcoin's futures on CMEG were only introduces at the end of 2017, so the data only starts then
- ETFs are assumed to have the same trading intensity. This allows us to only take a sample (the biggest ETF) and calculate the intensity based on it. If the assumption holds, this gives a valid estimate. This does not truly hold in reality, but the bias is limited.
- For S&P 500 ETF no data from Bolsa De Valores De Colombia and Tel Aviv Stock Exchange was fetched. Together they make about ~3% of the total volume (source: ChatGPT). So, the trading intensity for S&P 500 is underestimated. The unbiased estimate will however not change the statement of the figure

- For NASDAQ, volume from Tel Aviv Stock Exchange is missing. This leads to an underestimation of trading intensity by around 4%. Same as with S&P 500, the statement of the figure does not change.
- Gold's volume is also underestimated by around 5-10%. This again does not change the statement the figure makes
- Gold begins in 2013 due to CMEG beginning to post their monthly reports then
- CMEG's MICRO GOLD is not counted. Reason: counting it would introduce an additional level of complexity and is not expected to change overall interpretation of the trading intensity drastically
- Same as with gold, oil's volume is also underestimated by 2-10%.
- Oil's monthly production is aggregated from one-day production. More in the notebooks and data source.
- Oil's trading intensity begins in 2013, because CMEG started posting their monthly reports then

References

- ⇨ **Data sources:** `DATA_REFERENCE.md`
- ⇨ **Code to create figures 1-7:** `phase1/analysis.ipynb`
- ⇨ **Code to create figure 8-10:** `phase2/analysis.ipynb`
- ⇨ **Project planning** (high- & low-level designs): `planning/**`
- ⇨ **Project tree:** `PROJECT_TREE.md`